

CO2 - AT1

Analytical Problem Solving

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Sub code: ITA0502

Sub name: Computer Vision

① Gray Level Transformation

Q Explain the Purpose of grey level transformation in image formation?

A) Grey Level Transformation is a Point Processing technique in which every Pixel intensity of an image is modified according to a mathematical function.

Purpose:-

- Improves the brightness of a image
- Enhances the contrast b/w objects and background.
- Highlights important image details.

Formula:-

$$S = 255 - T$$

T = original Pixel intensity

S = Transformed Pixel intensity

⑥ Apply the negative transformation.

Given,

$$T = 50, 100, 150, 200$$

$$S = 255 - T$$

$$\text{For } T = 50$$

$$S = 255 - 50 \\ = 205$$

$$\text{For } T = 100$$

$$S = 255 - 100 \\ = 155$$

$$T = 150$$

$$S = 255 - 150 \\ = 105$$

$$T = 200$$

$$S = 255 - 200 \\ = 55$$

original Pixel	Transformed Pixel
50	205
100	155
150	105
200	55

- Dark pixels become bright
- Bright Pixels become dark
- Image colors are inverted.

② Histogram Equalization

① Explain why histogram equalization is used?

A) Histogram equalization is an image enhancement technique used to improve image contrast by redistributing Pixel intensity values over the entire available intensity range.

Given Histogram:-

Intensity	Frequency
0	4
1	6
2	10
3	6

Cumulative Freq

$$N_1 = 4$$

$$N_2 = 4 + 6 = 10$$

$$N_3 = 10 + 6 + 4 = 20$$

$$N_4 = 10 + 6 + 6 + 4 = 26$$

$$N = 26$$

⑥ compute CDF

$$CDF = \frac{\text{Cumulative Frequency}}{26}$$

Intensity	CDF frequency
0	$4/26 = 0.154$
1	$10/26 = 0.385$
2	$20/26 = 0.769$
3	$26/26 = 1.000$

pdf

⑦ Determine Equalized intensities

Given $L = 4$.

Transformation Formula =

$$S = (L-1) \times CDF$$

$$S = 3 \times CDF$$

$$i = 0 \Rightarrow 3 \times 0.154 = 0.462 \approx 0$$

$$i = 1 \Rightarrow 3 \times 0.385 = 1.155 \approx 1$$

$$i = 2 \Rightarrow 3 \times 0.769 = 2.307 \approx 2$$

$$i = 3 \Rightarrow 3 \times 1 = 3$$

Interpretation:-

- Pixel values become more evenly distributed.
- Contrast improves.
- Dark and brightness regions becomes more distinguishable.

3) Gaussian Smoothing

(a) Explain why Gaussian Filtering is Preferred over averaging filtering.

⇒ Gaussian Filtering Performs weighted averaging where pixels nearer to the center receive greater weight.

(b) Apply the Gaussian kernel to image region

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

A) (Multiplying Both $\Rightarrow (1 \times 10) + (2 \times 20) + (1 \times 10) + (2 \times 20) + 4(40) + 2(20) + (1 \times 10) + (2 \times 20) + (1 \times 10)$

$$\Rightarrow 10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10 \Rightarrow 60 + 240 + 60 \Rightarrow 360$$

$$\Rightarrow \frac{1}{16}(360) \Rightarrow \frac{360}{16} = 22.5 \approx 23$$

∴ New Center Pixel = 22.5

4) Edge Detection using Sobel operation.

(a) Explain the concept and Purpose of edge detection

⇒ Edge detection identifies locations where Pixel intensity changes detection rapidly.

⇒ Sobel operator computes the image gradient using convolution masks.

(b) Apply Horizontal Sobel mask.

$$\text{Mask} = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\text{Image} = \begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

A) Multiplying Both

$$\Rightarrow (-1 \times 10) + (-2 \times 10) + (-1 \times 10) + 0 + 30 + 60 + 30$$

$$\Rightarrow -10 - 20 - 10 + 0 + 120$$

$$\Rightarrow -40 + 120 = 80$$

Gradient = 80

Interpretation:-

⇒ A large positive gradient indicates a strong horizontal edge.

⇒ Brightness increases from top to bottom.

⇒ The sobel operator successfully detects the edge.

5 Contrast stretching

(a) Purpose:-

Contrast stretching increases the dynamic range of image intensities.

(b) Linear contrast stretching:-

Original range = 50 - 150

New range = 0 - 255

$$\text{Formula} = S = \frac{(r - r_{\min})}{r_{\max} - r_{\min}} \times 255$$

$$r_{\min} = 50, \quad r_{\max} = 150$$

Given $r = 50, 100, 150$.

$$r = 50 \Rightarrow S = \frac{(50 - 50)}{150 - 50} \times 255 = \frac{0}{100} \times 255 = 0$$

$$r = 100 \Rightarrow S = \frac{100 - 50}{150 - 50} \times 255 = \frac{50}{100} \times 255 = 127.5 \approx 128$$

$$r = 150 \Rightarrow S = \frac{150 - 50}{150 - 50} \times 255 = \frac{100}{100} \times 255 = 255$$

Final Table

Original Intensity	final intensity.
50	0
100	128
150	255

Interpretation:-

- minimum intensity becomes 0 (black)
- maximum intensity becomes 255 (white)
- mid level intensity becomes 128 (gray)
- The intensity range is expanded from 50-150 to 0-255, improving image contrast